
How Wrong Can Your Simulator Be? Auditing, Tolerating, and Updating a Measured Forward Model on a Real Instrument

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Abstract

1 A learned restoration model for degraded macro X-ray fluorescence (MA-XRF)
2 scans of a painting is trained on a forward simulator whose every ingredient
3 (geometry, per-line gains, noise law) is measured on the instrument; we treat
4 the simulator’s imperfection as the object of study. An ensemble trained across
5 simulators drawn within the measured calibration uncertainty separates simulator
6 uncertainty from initialisation variance ($2.8\times$ wider than a fixed-simulator control
7 on the real scan, 86 % of its variance attributable to the simulator), yields a near-
8 calibrated band on the real scan (coverage 0.61/0.87/0.95 at $1/2/3\sigma$), and its mean is
9 the first learned restoration in our study that does not lose correlation to the physics
10 inverse on the real scan. A ladder of 18 deliberately broken training simulators
11 shows the learned refinement tolerates large noise and gain errors but fails on
12 geometry: 0.5 px of registration error already hurts, and 1 px makes the network
13 worse than the physics it refines. A blind diagnostic battery, thresholded on the
14 calibration uncertainty itself, identifies every defect that causes material damage,
15 and on the real scan rediscovers, unsupervised, the resampling-blur mismatch that
16 had originally broken our training pipeline. Used as ABC summaries, the same
17 statistics turn one real scan into a posterior over the simulator: the blur defect is
18 rejected (prior probability 0.5, posterior 0.000) and a posterior-trained ensemble
19 is the most accurate restoration in the study, with a 27 % narrower band than the
20 prior ensemble. Finally, a 180-case degradation grid with four classical inpainting
21 controls shows where learning pays: everywhere on measured pixels, nowhere
22 inside missing regions, where a biharmonic fill wins in 16 of 16 cells. All results
23 are anchored on real measurements from one instrument and one painting; all
24 negative results are reported.

25 1 Introduction

26 Machine learning for experimental science usually trains on simulations because ground truth is
27 scarce, and the simulator is never exactly the instrument; the common response is to validate it once
28 and trust it. This paper takes the opposite view on a small but fully measured problem: we ask *how*
29 imperfect a validated simulator still is, what that imperfection costs a network trained on it, whether
30 it can be detected and attributed blind, how to represent it honestly in predictive uncertainty, and
31 whether a single real measurement can reduce it.

32 Our testbed is the restoration of geometrically degraded MA-XRF element maps of an oil painting
33 (Alfeld et al., 2013): a scan acquired at a tilt must be registered, gain-corrected and denoised back
34 into the frontal frame. The forward model is not designed but measured (Section 2), which gives

us something rare: a per-component calibration uncertainty for every knob of the simulator. Our contributions:

- **Uncertainty that knows the simulator is imperfect.** Deep ensembles (Lakshminarayanan et al., 2017) trained across simulator draws within the measured calibration uncertainty, against a fixed-simulator control that isolates initialisation variance (Section 3): domain randomization (Tobin et al., 2017) with measured ranges, evaluated as uncertainty quantification.
- **A defect-tolerance ladder and a blind audit.** One network per deliberately broken simulator quantifies which defects hurt; a battery of summary statistics, thresholded on the calibration uncertainty, identifies which component is broken without ground truth (Section 4). Every defect that causes material damage is visible to the audit.
- **Closing the loop.** The audit statistics are ABC summaries (Beaumont et al., 2002; Cranmer et al., 2020): one real scan yields a posterior over the simulator’s knobs, a posterior-trained ensemble, and a posterior predictive check that names precisely what the simulator still lacks (Section 5).
- **A regime map against classical baselines.** A 180-case grid over tilt, missing data and dose, with four classical inpainting controls, locates the frontier where learning pays (Section 6); inside missing regions the answer is unflattering and we report it as such.

2 Setup: a measured forward model, and how it burned us

Instrument and data. Two silicon drift detectors (70 mm² mounted horizontally and 25 mm² at 40° from horizontal, 12.5 µm Be windows) and a tube at 37 kV / 40 µA scan an oil painting; per-pixel spectra are fitted into net-count maps of 8 element lines (Ca, Ti, Fe, Cu, and four Pb L lines), summed over the detectors. Scans: two frontal, F_1 and F_2 (60 × 120 px, seven days apart, same geometry), and one real tilted, T (45 × 80, mounted at 7.7°). F_1 is the only training source; F_2 and T are reserved for evaluation and are never seen in training.

Forward model. Every ingredient is measured, not modelled: the warp is the affine registration fitted between T and F_1 ; the per-line tilt gains are the measured level response at the reference mounting, extrapolated linearly in angle; the noise is Poisson-consistent with a per-line constant, $\text{Var}(m) = k_{\text{el}}m$ with $k_{\text{el}} = 4.2\text{--}9.0$ calibrated from the $F_1\text{--}F_2$ difference. A fidelity gate holds on all 8 lines: $r(\text{sim}, T) = 0.89\text{--}0.98$, within 0.02 of the $F_1\text{--}F_2$ repeatability ceiling, and levels within 0.7 %. A residual U-Net (Ronneberger et al., 2015) (469k parameters, zero-initialised head, so the untrained network *is* the physics inverse) refines the deterministic inversion; training pairs are simulated from F_1 at random angles, doses and dropout blocks (Appendix A).

The blur lesson. Our first training simulator emulated the tilted acquisition by bilinearly warping the frontal map. It passed the fidelity gate, yet the network trained on it *over-sharpened* the real scan (contrast ratio up to 1.05, correlation below the physics inverse on 8/8 lines). The cause: a real tilted scan is a direct measurement and carries no resampling blur, so the simulated restoration inputs carried two interpolation blurs where reality has one. Sampling the source with a cubic spline at the exact warp positions (v2) fixed the training distribution. The lesson behind everything below: *validated as an emulator is not validated as a training-data generator*, and the failure was invisible on simulated tests.

Perturbable simulator. We expose the simulator’s components as knobs θ : a noise-constant scale, a tilt-gain slope scale, per-line gain offsets, the assumed mounting angle, registration shift and rotation, the interpolation order (cubic vs the v1 bilinear), and, in Section 5, per-line noise scales. Each knob carries a measured 1σ calibration uncertainty (noise log-sd 0.33 from the session transfer of k ; gain slope 20 %, the mounting angle being an upper bound; registration 0.1 px, 0.3° from the spread of per-element fits). Perturbations apply only to the training-data simulator; the test-time inverse stays nominal, as in practice.

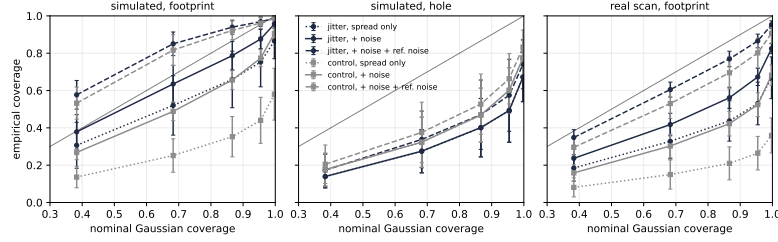


Figure 1: Coverage of $|\text{mean} - \text{truth}| \leq z\sigma$ vs the Gaussian target, jitter (navy) and control (grey); dotted: spread only, solid: plus propagated noise, dashed: plus reference-map noise (upper bound on simulated cases, the correct band on the real scan). Bars: sd over lines and cases. Left: 32 simulated cases (footprint); middle: inside a 14×20 hole; right: the real scan.

83 3 Uncertainty that knows the simulator is imperfect

84 We train two ensembles of $N=12$ members each: a *jitter* ensemble, member i trained on a simulator
 85 drawn from the calibration uncertainty, and a *control* ensemble with the nominal simulator and the
 86 same twelve seeds, the standard deep-ensemble baseline (Lakshminarayanan et al., 2017). By con-
 87 struction the excess spread of the jitter ensemble is the variance attributable to simulator uncertainty.
 88 The predictive band adds measurement noise propagated through 8 re-simulations per case plus,
 89 on the real scan, the calibrated noise of the reference map ($k_{\text{el}} \cdot \text{truth}$), the reference being itself a
 90 measurement.

91 **The ensemble knows where the simulator is uncertain.** On the footprint the jitter ensemble spreads
 92 $2.2\times$ (simulated) and $2.8\times$ (real scan) wider than the control; 62 % and 86 % of its variance is
 93 attributable to the simulator draw. The simulator share grows monotonically with tilt, 104 to 148
 94 counts rms from 8° to 25° , the direction of the gain extrapolation from the single 7.7° calibration;
 95 spatially it follows edges and levels, where registration and gain errors show, while the control spread
 96 lives only in dropout holes (Appendix Fig. 5).

97 **Calibration.** With spread alone both ensembles under-cover (real scan, $z=2$: 0.53 jitter, 0.26 control,
 98 target 0.95), as expected when measurement noise dominates. With the full band the jitter ensemble
 99 covers 0.61/0.87/0.95 at $z = 1/2/3$ on the real scan against 0.53/0.80/0.91 for the control (Fig. 1);
 100 on simulated cases it is bracketed between 0.64 and 0.85 at $z=1$ because the reference maps are
 101 themselves noisy. The jitter spread also ranks the real-scan errors better (Spearman 0.31 vs 0.26,
 102 AUSE (Ilg et al., 2018) 0.33 vs 0.38).

103 **Where it fails.** Inside a 14×20 dropout hole neither ensemble is calibrated ($z=2$ coverage 0.49/0.60
 104 against rms error 940 counts vs spread 320–410), the spread does not rank the error (Spearman $\approx$
 105 0.1), and the jitter ensemble is *narrower* than the control (ratio 0.85): all members share the same
 106 training-image prior and agree on the same wrong fill. Simulator uncertainty is not prior uncertainty,
 107 and missing data exposes the difference.

108 **Jitter training costs nothing on the real scan.** The jitter-ensemble mean matches or exceeds
 109 the physics inverse in r on 5/8 lines (mean 0.9527 vs 0.9526, the other lines within 0.002) while
 110 restoring contrast (cv ratio 0.979 vs 0.947), whereas the single nominal network and the control mean
 111 lose r on 8/8 lines (0.9485, 0.9496). Calibration-posterior randomization is a physically grounded
 112 augmentation against exactly the residual mismatch of the real scan, at a price of 0.003 in r on
 113 simulated cases (there the nominal simulator is true).

114 4 Auditing the simulator: tolerance and blind diagnostics

115 **Tolerance ladder.** One network per deliberately broken simulator, 18 rungs across six families, all
 116 with one shared seed so that rung differences are the simulator; the 12 control members provide the
 117 seed band ($\Delta r \in [+0.082, +0.090]$ on the simulated footprint, $[-0.005, -0.002]$ on the real scan).
 118 The result (Fig. 2) is asymmetric: the noise constant can be wrong by $4\times$ either way, the gain slope
 119 doubled or removed, the angle belief off by 5° , all without leaving the band. Geometry is different:
 120 a 0.5 px registration shift, five times the calibration uncertainty but half a pixel, already leaves the

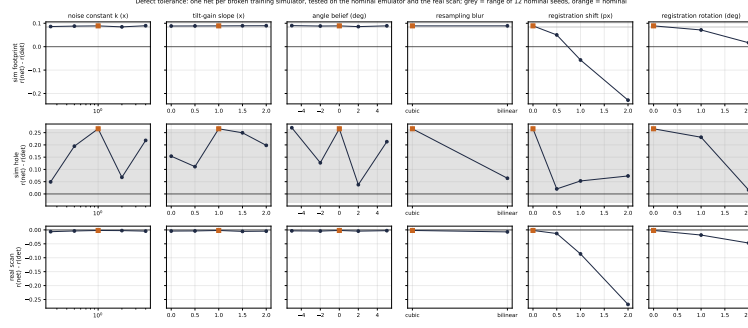


Figure 2: Defect tolerance: $r(\text{net}) - r(\text{physics})$, one network per broken training simulator, on nominal simulated cases (top: footprint, middle: hole) and the real scan (bottom). Grey band: 12 nominal seeds; orange: nominal. Geometry is the cliff; the blur defect is visible only on the real scan.

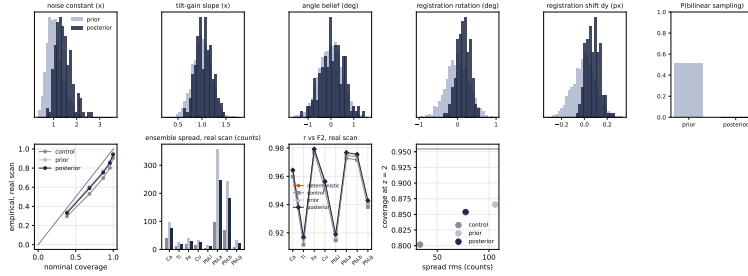


Figure 3: Top: prior (light) and posterior (navy) knob marginals from rejection ABC on one real scan; the v1 bilinear sampling drops from 0.5 to 0.000. Bottom: real-scan calibration, per-line spread and r , and the coverage-vs-width tradeoff of the three ensembles.

band (four seeds: -0.013 to -0.005 on the real scan); at 1 px the network is worse than the physics inverse it refines (-0.057 simulated, -0.086 real), at 2 px it destroys the restoration (-0.27 real); a 2° rotation costs -0.047 . The blur rung replays Section 2 as one point on a curve: invisible on simulated tests, on the real scan it leaves the band for all four seeds, with the table’s only contrast overshoot (1.013).

Blind diagnostics. Can the broken component be identified from one scan, without truth? We compare measurement to simulation with nine per-line statistics (level, contrast, mid- and high-band power ratios, a single-map noise-constant estimate, signed sub-pixel offsets, a differential rotation estimate, a gain-template projection). The null is the crucial choice: with noise-only pairs everything fires on the real scan (a 0.3° registration residual gives $z \approx 60$), so we build it from pairs drawn *within the calibration uncertainty*: a flag means the discrepancy exceeds what the simulator itself claims to know. With a rule fixed before the experiment, the battery identifies blur 10/10, shifts down to 0.5 px 30/30, rotations of $1-2^\circ$ 20/20 and noise scalings $\times 0.25/2/4$ 30/30, with 4/48 false alarms; gain-family defects within the calibration uncertainty are invisible by construction (0/80 pre-registered; a post-hoc template recovers only extremes; gain vs angle is degenerate at one tilt). Crossing audit and ladder gives the payoff matrix: *every rung with material damage is visible, and every invisible rung is harmless* (one formal exception at 0.0002, the band’s own resolution). On the real scan the battery, blind, flags our v1 simulator and the validated bilinear emulator as “blur” (high-frequency power ratio 1.7–2.1) and leaves the v2 simulator with one finding: a residual noise excess (k ratio 1.19), i.e. it rediscovers the blur lesson and names the next problem (Appendix Fig. 7).

5 Closing the loop: a posterior over the simulator from one scan

The audit statistics support likelihood-free inference (Cranmer et al., 2020): rejection ABC (Beaumont et al., 2002) over the simulator’s knobs, prior = the calibration jitter of Section 3 plus the v1 bilinear

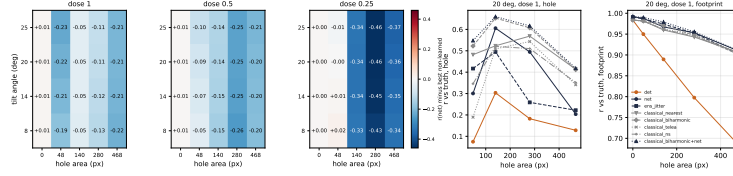


Figure 4: Left: $r(\text{net})$ minus the best non-learned candidate (physics or best classical fill) inside the hole, per dose; blue = the non-learned candidate wins. Right: r against hole area at 20° , dose 1, all candidates, inside the hole and on the footprint.

144 sampling at probability 0.5 (the loop must reject our own mistake); 3000 draws, whitened RMS- z
145 distance, closest 5 % accepted (2/10 % agree on every direction).

146 **Posterior.** The bilinear defect is rejected completely (posterior 0.000); the posterior concentrates
147 on a noisier instrument (noise scale 1.43 ± 0.35) with a small registration residual ($+0.15^\circ \pm 0.17$,
148 $+0.08 \pm 0.06$ px), gains at calibration. The posterior predictive check stays honest: no draw is
149 fully consistent (0/12), the verdict is uniformly “noise”, and its decomposition shows a *per-line*
150 residual (real-to-simulated variance ratios 0.9–4.6 across lines) that a global noise knob cannot
151 express. Adding the named knob (per-line noise scales, round 2) moves the marginals in the measured
152 direction but barely deepens the fit (best whitened distance 2.02 vs a null floor of 0.95 ± 0.24) and
153 the check still rejects: the loop converges not to perfect but to a precise statement of what is missing
154 (unmodelled flat-field, scatter, dwell drift).

155 **Posterior ensemble.** Twelve networks retrained on accepted draws form the most accurate restoration
156 in the study on the real scan: $r = 0.9539$ against 0.9527 (prior), 0.9526 (physics), 0.9496 (control),
157 at or above the physics inverse on 7/8 lines, with the smallest bias (1.07 %) and a 27 % narrower
158 band than the prior ensemble (spread 77 vs 106 counts) at coverage 0.85/0.94 at $z = 2/3$ (prior
159 0.87/0.95). By per-pixel Gaussian NLL, a proper score (Gneiting and Raftery, 2007), the wider prior
160 band remains marginally better (6.45 vs 6.58 nats; control 7.27), the gap sitting exactly on the lines
161 whose noise the knob family cannot express: one measurement bought accuracy, bias and band width,
162 and the score prices the misspecification that remains.

163 6 When does learning pay? A regime map with classical controls

164 We sweep 180 cases ($4 \text{ angles} \times 5 \text{ holes} \times 3 \text{ doses} \times 3 \text{ seeds}$, all from F_2) and compare physics, the
165 nominal network, the jitter-ensemble mean, and four classical fills applied in the tilted frame (nearest,
166 biharmonic (van der Walt et al., 2014), Telea (Telea, 2004), Navier–Stokes (Bertalmio et al., 2001))
167 through the same physics inverse. On measured pixels the learned candidates beat every alternative
168 in every cell (no hole: 0.985 vs 0.977 physics; 14×20 hole footprint: 0.952 ensemble vs 0.937 best
169 classical vs 0.781 physics), with a margin that grows with hole size (they also repair the warp blur
170 around the hole). Inside the hole the ranking inverts: the biharmonic fill beats the network in 16/16
171 cells (0.467 vs 0.415 at $20^\circ/14 \times 20$), and at quarter dose the learned fills collapse (0.03 and -0.01)
172 while the classical fill is dose-proof (0.455). The only learned candidate that wins every hole cell is
173 the hybrid, classical fill then network, by 0.005–0.03: the network adds sharpening, not a better fill.
174 Hole columns are angle-independent (the measured warp does not scale with angle); what matters is
175 hole size and dose. On this instrument the learned prior buys contrast, noise, neighbourhood repair
176 and calibrated uncertainty, not missing content (Fig. 4).

177 7 Limitations and outlook

178 All of this is one painting, one instrument, one real tilted scan; we claim the auditable chain, not
179 universal numbers. The calibration sigmas are lower bounds (the one-sided gain-slope uncertainty is
180 symmetrised), and the jitter is only as honest as they are. Ensembles have 12 members; rungs use one
181 seed, the two borderline rungs four. Rejection ABC with 9×8 summaries and 22 round-2 parameters
182 is at its resolution edge; SMC is the upgrade. The hole failure suggests training on classically filled
183 holes; uncertainty-guided acquisition (pilot in Appendix Fig. 8) is the route to less irradiation of
184 heritage objects, our main foreseen societal benefit. Code and result tables will be released.

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A Training and evaluation details

Data and protocol. Frontal maps are 60×120 , the tilted frame 45×80 ; the tilted footprint covers 3466 of the 7200 frontal pixels. Training pairs are simulated from F_1 only: random tilt angle in $[4^\circ, 25^\circ]$, random flips, 25 % low-dose samples ($\times 0.35$ – 1.0), 30 % of samples with 1–2 zeroed dropout blocks of 8–17 px in the tilted frame, recorded in a validity channel. The network input is the deterministic physics inverse of the simulated tilted scan (8 element channels, per-line normalisation fixed from F_1), plus validity and angle channels; the target is the frontal map. Four fixed 15×30 spatial blocks of the frontal frame are excluded from the training loss and serve early stopping. F_2 and the real scan T are never used in training; all final numbers are computed against F_2 as reference, on the footprint and, where applicable, inside the dropout hole. Metrics: Pearson r , SSIM, level bias in percent, and the contrast ratio $CV(\text{candidate})/CV(\text{reference})$.

Architecture and optimisation. Residual U-Net, three resolution levels, base width 32, Group-Norm and SiLU, 469k parameters, 1×1 zero-initialised head (the untrained model equals the physics inverse). Adam (Kingma and Ba, 2015) with learning rate 10^{-3} , batch 8, at most 1500 steps, validation every 50 steps on a fixed simulated set (8 angles $\times$ 3 replicates), early stopping after 10 validations without improvement, plateau learning-rate decay. One training run takes about 10 minutes on a 12-core CPU workstation; no GPUs were used anywhere. The study trains 60 networks in total (12 jitter, 12 control, 18 ladder rungs, 6 extra-seed variants, 12 posterior members), about 10 CPU-hours, plus inference-only sweeps.

229 **Seeds and error bars.** The two ensembles share their twelve seeds, so jitter-vs-control differences
 230 are attributable to the simulator draw; every ladder rung uses the seed of one control member, and the
 231 12 control members define the initialisation band shown in Fig. 2; the two borderline rungs (0.5 px
 232 shift, blur) were re-trained with three extra seeds each and their real-scan damage holds for every
 233 seed. Coverage error bars are standard deviations over lines and cases. The aleatoric band term uses
 234 8 noise re-simulations per case; the reference-noise term uses the calibrated k_{el} .

235 B Additional figures

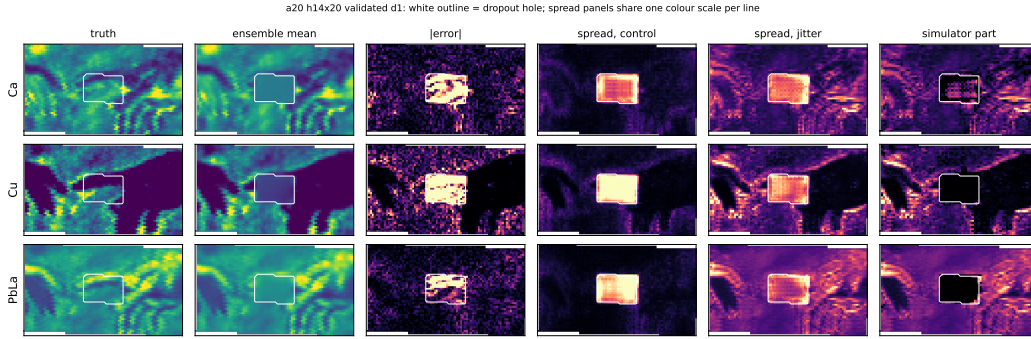


Figure 5: Spread maps on the harsh case (20° , 14×20 hole, white outline), three lines. Columns: reference, jitter-ensemble mean, absolute error, control spread, jitter spread, and the simulator part $\sqrt{\max(\sigma_{\text{jitter}}^2 - \sigma_{\text{control}}^2, 0)}$; the three spread panels of a line share one colour scale. The control spread lives in the hole; the jitter spread follows the painting's structure; inside the hole the jitter ensemble is narrower than the control (dark).

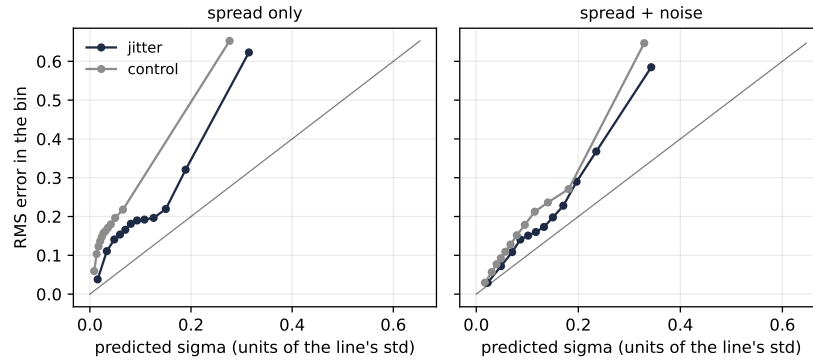


Figure 6: Binned reliability: RMS error against predicted σ in twelve σ -quantile bins, pooled over simulated cases and lines in units of each line's standard deviation; diagonal = calibrated. Left: ensemble spread only; right: spread plus propagated noise.

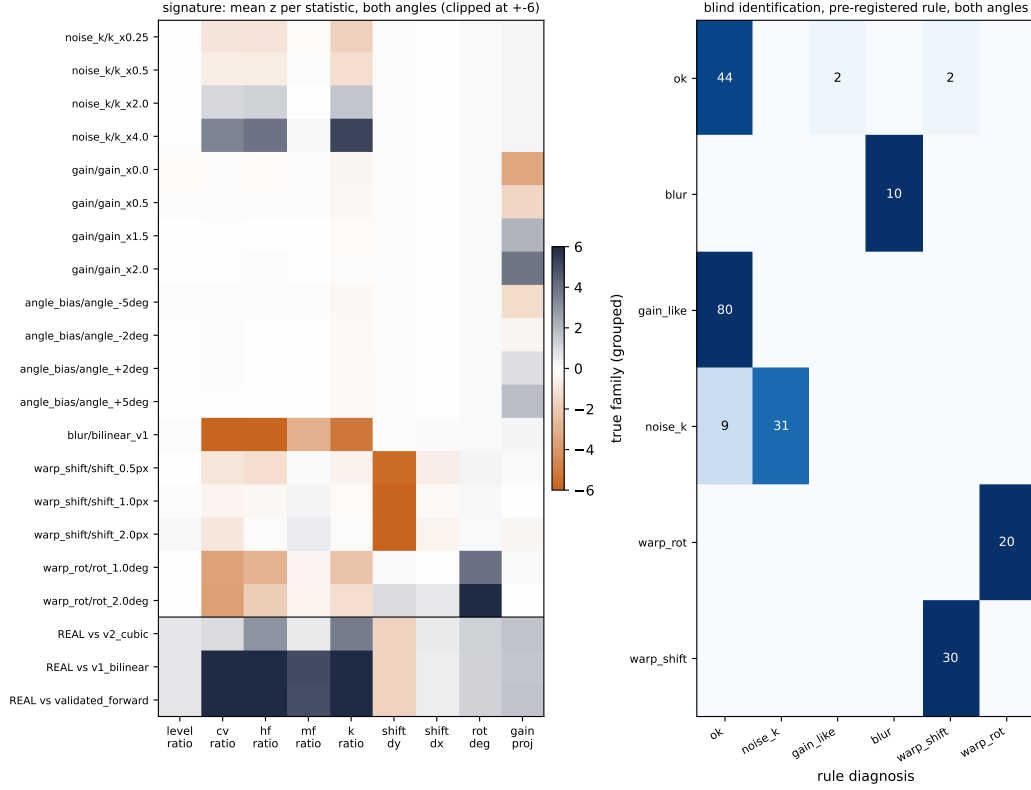


Figure 7: Blind diagnostics. Left: signature of every defect rung, the mean z of each statistic against the calibration-uncertainty null (both test angles pooled, clipped at ± 6); the bottom block is the real scan against three simulators (v2 cubic, v1 bilinear, validated bilinear emulator). Right: identification counts of the pre-registered rule, true family (grouped) versus diagnosis.

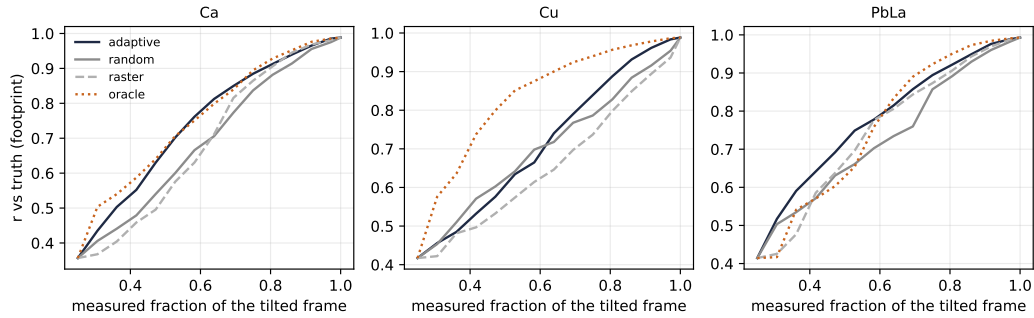


Figure 8: Uncertainty-guided acquisition pilot: r against the measured fraction of the tilted frame when 5×5 tiles are acquired in the order of the jitter-ensemble spread (adaptive), at random, in raster order, or by the true error (oracle, not realisable); 8 tiles per step from a 25 % start, mean of three simulated cases. Adaptive reaches $r \geq 0.90$ (mean over Ca, Cu, PbLa) after measuring 79 % of the frame, against 85 % for random or raster order and 72 % for the oracle; for $r \geq 0.95$ the fractions are 88 %, 93 % and 82 %.

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Justification: Sections 2 to 6 define the measured forward model, the knob set with its calibration sigmas, both ensembles, the defect ladder, the diagnostic battery with its null model and pre-registered decision rule, and the ABC protocol; Appendix A gives the data protocol, architecture, optimisation, seed policy and compute.

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